



**THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED**

Q.1. Write short answer of the following questions: (10x3=30)

- (i)- What is meant by operators and state vectors in quantum mechanics? Also explain dimensionality and basis.
- †(ii)- What is meant by discrete and continuous spectrum? Explain it by considering the example of a general potential.
- (iii)- State any three postulates of quantum mechanics.
- (iv)- If  $x$  and  $p$  are position and momentum variables and

$$\hat{A} = \frac{1}{\sqrt{2m\hbar\omega}}(m\omega x + ip)$$

Show that  $[\hat{A}, \hat{A}^\dagger] = 1$ .

- (v)- Find out classical amplitude for a particle undergoing simple harmonic motion if its energy is  $\hbar\omega/2$ .
- (vi)- A particle is in state given by wave function,

$$\psi(x, t) = e^{-ax^2 - bt}$$

where  $a$  and  $b$  are constants.

Is it a stationary state? Find expectation values of position and momentum operators.

- (vii)- Show that if an operator  $\hat{A}$  has an eigen value  $\alpha \neq \alpha^*$ , then  $\hat{A}$  is not Hermitian.
- (viii)- Given that in spherical polar coordinates  $\hat{L}_z = -i\hbar \frac{\partial}{\partial \phi}$ , show that

$$[\hat{L}_z, \sin\phi] = -i\hbar \cos\phi$$

- (ix)- The operator for  $z$ -component of angular momentum is  $\hat{L}_z = -i\hbar \frac{\partial}{\partial \phi}$ . Determine whether or not  $\sin m\phi$  is its eigen function?
- (x)- State results of Stern-Gerlach experiment.

Q.2. Solve the following.

(3x10=30)

- (i)- State and prove generalized uncertainty relation (Uncertainty principle for operators).
- (ii)- Differential equation for simple harmonic oscillator obtained after solving Schrodinger wave equation is given by,

$$H''(z) - 2zH'(z) + (\lambda - 1)H(z) = 0$$

Solve this equation to find energy eigen values and wave functions of simple harmonic oscillator.

- (iii)- Discuss the significance of ladder operators  $\hat{L}_\pm$ . Show that they have following property.

$$\hat{L}_+ \psi(l, m) = \sqrt{l(l+1) - m(m+1)} \hbar \psi(l, m+1)$$

✓  
1s

**3.2: Operators and state vectors:** In quantum mechanics the physical state of a system is represented by elements of Hilbert space  $\mathcal{H}$  called state vectors. The meaning of a vector in quantum mechanics is that it is independent of basis in which it is expanded. Ket vectors represent the system completely. By knowing  $|\psi\rangle$ , we can learn all about its amplitude in all possible representations. If  $|\psi\rangle$  and  $|\phi\rangle$  belong to same vector space, products of type  $|\psi\rangle|\phi\rangle$  and  $\langle\psi|\langle\phi|$  are not allowed because these are neither ket nor bra. For each linear operator  $\hat{A}$ , we can write

$$\hat{A} = I\hat{A}I = \left( \sum_{i=1}^{\infty} |\psi_i\rangle\langle\psi_i| \right) \hat{A} \left( \sum_{j=1}^{\infty} |\psi_j\rangle\langle\psi_j| \right) = \sum_{i,j=1}^{\infty} A_{ij} |\psi_i\rangle\langle\psi_j|$$

where  $A_{ij}$  are matrix elements of operator  $\hat{A}$ :

$$A_{ij} = \langle\psi_i|\hat{A}|\psi_j\rangle$$



**Dimensionality and basis:** - A set of linearly independent vectors  $\{|\psi_1\rangle, |\psi_2\rangle, \dots, |\psi_n\rangle\}$  in a vector space  $V$  such that every vector of  $V$  can be expressed as a linear combination of vectors  $|\psi_1\rangle, |\psi_2\rangle, \dots, |\psi_n\rangle$ ; is called a basis for vector space  $V$  and the vectors  $|\psi_1\rangle, |\psi_2\rangle, \dots, |\psi_n\rangle$  are said to span vector space  $V$ . The number of basis vectors of a vector space  $V$  is called dimension of that space. If this number is finite, we have finite dimensional space otherwise infinite.

2s

orthogonal.  $\psi$  state

**3.4: Discrete and continuous spectra:** Bound state problems are those in which the motion of particle is restricted by external forces (potential energy) to a particular region of space. For example the one dimensional motion of a particle attached to a fixed centre i.e. linear harmonic oscillator and infinite square well potential. Similarly the hydrogen atom in which one electron is moving around the positively charged nucleus is of special importance in quantum mechanics. For bound states, set of eigen values and eigen functions are discrete sets.

Unbound state problems are those in which the motion of particle is not restricted by external forces to a particular region of space e.g. free particle. For unbound states, set of eigen values and eigen functions are continuous sets.

To understand concept of discrete and continuous spectra, consider a general potential shown in fig 3.2.

In fig 3.2., motion of particle is bounded between points  $x_1$  and  $x_2$  when the energy of particle lies between  $V_0$  and  $V_1$ . The states corresponding to this energy range are called bound states. The motion of particle is unbounded for energy ranges  $V_1 < E < V_2$  and  $E > V_2$ . The states corresponding to this energy range are called unbound states. The spectrum of bound states is discrete and non-degenerate. Unbound states cannot be normalized.



3s

**2.4: Postulates of quantum mechanics:** In Schrodinger formulation of quantum mechanics, a complex variable quantity  $\psi(x, y, z, t)$ , called the wave function, was introduced to represent the wave associated with a particle in motion. Quantum mechanics is based on certain postulates, which are sufficient to describe a single particle system.

**Postulate I:** Corresponding to every observable, there exists an operator. The only measurable values of a physical observable are the various eigen values of corresponding operator. In classical mechanics, physical observable are represented by dynamical variables such as energy, angular momentum etc. In quantum mechanics, we use their corresponding operators. For example, position operator is represented as  $\hat{x}$ , potential energy operator as  $\hat{V}(x)$  and momentum operator:

$$\hat{p}_x = -i\hbar \frac{\partial}{\partial x} \text{ etc.}$$

**Postulate II:** For a system consisting of a particle moving in a field of conservative force, there is an associated complex wave function  $\psi(x, y, z, t)$ , where  $x, y, z$  are space coordinates and 't' is time. This function enables us to obtain a description of the behavior of the system consistent with uncertainty principle.

**Postulate III:** The wave function  $\psi(x, y, z, t)$  and its partial derivatives  $\frac{\partial \psi}{\partial x}, \frac{\partial \psi}{\partial y}, \frac{\partial \psi}{\partial z}$  must be finite, continuous and single valued for all values of  $x, y, z$  and 't'.

**Postulate IV:** The product  $\psi^*(x, y, z, t)\psi(x, y, z, t)$ , where  $\psi^*(x, y, z, t)$  is complex conjugate of  $\psi(x, y, z, t)$ , is always a real quantity. This product is called probability density and  $\psi^*(x, y, z, t)\psi(x, y, z, t)dV$  is probability that the particle will be found in this volume element  $dV$  at  $x, y, z$  and 't'. The total probability that the particle will be somewhere in space must be equal to unity, we have

$$\int_{-\infty}^{\infty} \psi^*(x, y, z, t)\psi(x, y, z, t)dV = 1$$

This condition is called normalization condition.

**Postulate V:** The average or expectation value of an observable quantity A with which an operator  $\hat{A}$  is associated, is defined by:

$$\hat{A} = \frac{\int_{-\infty}^{\infty} \psi^* \hat{A} \psi dV}{\int_{-\infty}^{\infty} \psi^* \psi dV}$$

But  $\int_{-\infty}^{\infty} \psi^* \psi dV = 1$ , so

$$\hat{A} = \int_{-\infty}^{\infty} \psi^* \hat{A} \psi dV$$



4s

If  $x$  and  $p$  are position and momentum variables and

$$\hat{A} = \frac{1}{\sqrt{2m\hbar\omega}} (m\omega\hat{x} + i\hat{p})$$

$$\hat{A}^\dagger = \frac{1}{\sqrt{2m\hbar\omega}} (m\omega\hat{x} - i\hat{p})$$

$$AA^\dagger = \left[ \frac{1}{\sqrt{2m\hbar\omega}} (m\omega\hat{x} + i\hat{p}) \right] \left[ \frac{1}{\sqrt{2m\hbar\omega}} (m\omega\hat{x} - i\hat{p}) \right]$$

$$= \frac{1}{2m\hbar\omega} (m\omega\hat{x} + i\hat{p})(m\omega\hat{x} - i\hat{p})$$

$$= \frac{1}{2m\hbar\omega} \{ m^2\omega^2\hat{x}^2 - im\omega\hat{x}\hat{p} + im\omega\hat{p}\hat{x} + \hat{p}^2 \}$$

$$= \frac{1}{2m\hbar\omega} \{ \hat{p}^2 + m^2\omega^2\hat{x}^2 - im\omega(\hat{x}\hat{p} - \hat{p}\hat{x}) \}$$

$$= \frac{1}{2m\hbar\omega} \{ \hat{p}^2 + m^2\omega^2\hat{x}^2 - im\omega(i\hbar) \}$$

$$= \frac{1}{2m\hbar\omega} \{ \hat{p}^2 + m^2\omega^2\hat{x}^2 + m\omega\hbar \}$$

$$= \frac{1}{\hbar\omega} \left( \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2 \right) + \frac{1}{2}$$

$$AA^\dagger = \frac{1}{\hbar\omega} \hat{H} + \frac{1}{2} \quad \text{--- (1)}$$

$$AA^\dagger - \frac{1}{2} = \frac{1}{\hbar\omega} \hat{H}$$

$$\hat{H} = \left( AA^\dagger - \frac{1}{2} \right) \hbar\omega \quad \text{--- (2)}$$



Similarly

$$A^t A = \frac{1}{\hbar\omega} \hat{H} - \frac{1}{2} \quad (2)$$

$$(A^t A + \frac{1}{2}) \hbar\omega = \hat{H}$$

$$\hat{H} = (A^t A + \frac{1}{2}) \hbar\omega$$

Subtracting (1) and (2)

$$AA^t - A^t A = \frac{1}{\hbar\omega} \hat{H} + \frac{1}{2} - \frac{1}{\hbar\omega} \hat{H} + \frac{1}{2}$$

$$AA^t - A^t A = \frac{1+1}{2} = \frac{2}{2} = 1$$

$$AA^t - A^t A = 1 \Rightarrow [A, A^t] = 1$$



5s

Solution:- For classical harmonic oscillator,

0.1 (v)

$$x = A_n \cos \omega t \text{ and } p = m \frac{dx}{dt} = -m\omega A_n \sin \omega t$$

Hence energy is,

$$E_n = \frac{p^2}{2m} + \frac{1}{2} m \omega^2 x^2 = \frac{m^2 \omega^2 A_n^2 \sin^2 \omega t}{2m} + \frac{1}{2} m \omega^2 A_n^2 \cos^2 \omega t$$
$$\Rightarrow E_n = \frac{1}{2} m \omega^2 A_n^2 \Rightarrow A_n^2 = \frac{2E_n}{m\omega^2} \Rightarrow A_n = \sqrt{\frac{2E_n}{m\omega^2}}$$

6s

showing that  $\hat{A}$  is not Hermitian operator.  
**Example 2.43:-** A particle is in state given by wave function,

$$\psi(x, t) = e^{-ax^2 - bt}$$

where  $a$  and  $b$  are constants.

a)-Is it a stationary state?

b)-Find expectation values of position and momentum operators.

c)-Does the result of (b) means that the uncertainty principle is violated?

Hint-a)-The probability density is given by,

$$P = \psi^*(x, t)\psi(x, t) = |\psi(x, t)|^2 = e^{-2ax^2 - 2bt}$$

Since probability density depends upon time, so the given state is not a stationary state.

[If wave function is  $\psi(x, t) = e^{-ax^2 - ibt}$ , then state function is stationary. Bonus]

b)-Expectation value of position operator is,

$$\langle \hat{x} \rangle = \int_{-\infty}^{\infty} \psi^*(x, t) \hat{x} \psi(x, t) dx = \int_{-\infty}^{\infty} e^{-ax^2 - bt} x e^{-ax^2 - bt} dx = \int_{-\infty}^{\infty} x e^{-2ax^2 - 2bt} dx$$

$$\langle \hat{x} \rangle = 0$$

Expectation value of momentum operator is,

$$\int_{-\infty}^{\infty} (\text{even function} \times \text{odd function}) dx = 0$$

$$\langle \hat{p}_x \rangle = \int_{-\infty}^{\infty} e^{-ax^2 - bt} \hat{p}_x e^{-ax^2 - bt} dx = \int_{-\infty}^{\infty} e^{-ax^2 - bt} \left( -i\hbar \frac{\partial}{\partial x} \right) e^{-ax^2 - bt} dx$$



$$\Rightarrow \langle \hat{p}_x \rangle = -i\hbar \int_{-\infty}^{\infty} e^{-ax^2-bt} \left( \frac{\partial}{\partial x} \right) e^{-ax^2-bt} dx = -i\hbar \int_{-\infty}^{\infty} e^{-ax^2-bt} (-2ax) e^{-ax^2-bt} dx$$

$$\Rightarrow \langle \hat{p}_x \rangle = -2ai\hbar \int_{-\infty}^{\infty} x e^{-2ax^2-2bt} dx = 0$$

c)-The result (b) does not mean that  $\Delta x \Delta p_x \geq \hbar$  is violated because  $\Delta x$  and  $\Delta p_x$  are defined by,

$$\Delta x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2} \quad \text{and} \quad \Delta p_x = \sqrt{\langle p_x^2 \rangle - \langle p_x \rangle^2}$$

i.e.  $\langle x \rangle$  and  $\langle p_x \rangle$  does not mean that  $\Delta x = 0$  and  $\Delta p_x = 0$ .

**Example 2.42:-** Show that if an operator  $\hat{A}$  has an eigen value  $a \neq a^*$ , then  $\hat{A}$  is not Hermitian.

**Hint:-** Let  $\psi$  be eigen function of operator  $\hat{A}$  (not Hermitian) corresponding to eigen value ' $a$ ', then

$$\hat{A}\psi = a\psi \rightarrow (a)$$

Taking complex conjugate,

$$(\hat{A}\psi)^* = a^* \psi^* \rightarrow (b)$$

Multiplying equation (a) with  $\psi^*$  and integrating,

$$\int_{-\infty}^{\infty} \psi^* \hat{A}\psi dx = \int_{-\infty}^{\infty} \psi^* a \psi dx \Rightarrow \int_{-\infty}^{\infty} \psi^* \hat{A}\psi dx = a \int_{-\infty}^{\infty} \psi^* \psi dx \rightarrow (c)$$

Multiplying equation (b) with  $\psi$  and integrating,

$$\int_{-\infty}^{\infty} (\hat{A}\psi)^* \psi dx = \int_{-\infty}^{\infty} \psi^* a^* \psi dx \Rightarrow \int_{-\infty}^{\infty} \psi^* \hat{A}^\dagger \psi dx = a^* \int_{-\infty}^{\infty} \psi^* \psi dx \rightarrow (d)$$

As  $\hat{A}$  is not Hermitian operator i.e.  $a \neq a^*$  so from equations (c) and (d), we get

$$\int_{-\infty}^{\infty} \psi^* \hat{A}^\dagger \psi dx \neq \int_{-\infty}^{\infty} \psi^* \hat{A}\psi dx$$

showing that  $\hat{A}$  is not Hermitian operator.



8s

## Chapter-05: Angular Momentum

23

Proof:- (i) -  $[\hat{L}_z, \cos\phi] = \hat{L}_z \cos\phi - \cos\phi \hat{L}_z$

Operating on a function  $\psi$ ,

$$[\hat{L}_z, \cos\phi]\psi = -i\hbar \frac{\partial}{\partial \phi} \cos\phi \psi - \cos\phi \left( -i\hbar \frac{\partial}{\partial \phi} \right) \psi$$

$$\Rightarrow [\hat{L}_z, \cos\phi]\psi = i\hbar \sin\phi \psi - i\hbar \cos\phi \frac{\partial \psi}{\partial \phi} + i\hbar \cos\phi \frac{\partial \psi}{\partial \phi} \Rightarrow [\hat{L}_z, \cos\phi]\psi = i\hbar \sin\phi \psi$$

As  $\psi$  is arbitrary function, so

$$[\hat{L}_z, \cos\phi] = i\hbar \sin\phi$$

✓ proved

Proceed similarly for part (ii).

9s

**Example 5.8:-** The operator for z-component of angular momentum is  $\hat{L}_z = -i\hbar \frac{\partial}{\partial \phi}$

**Determine whether or not  $\sin m\phi$  is its eigen function?**

**Solution:-** Now,

$$\hat{L}_z \sin m\phi = -i\hbar \frac{\partial}{\partial \phi} \sin m\phi \Rightarrow \hat{L}_z \sin m\phi = -i\hbar m \cos m\phi$$

This shows that  $\sin m\phi$  is not an eigen function of  $\hat{L}_z$ . However  $e^{im\phi}$  is an eigen function of  $\hat{L}_z$  corresponding to eigen value  $m\hbar$ .

**Example 5.9:-** Define ladder operators of  $\hat{L}_z$ .



## Stern Gerlach experiment

10s **Experimental results:** -Results of this experiment are listed below:

1. When experiment was performed with silver, a fairly sharp line was traced on plate when magnetic field was off. On establishing a highly inhomogeneous magnetic field, a double trace was obtained on either side of original line.
2. When experiment was performed with Cu, Au, H, Li, Na and K, then again a double trace was found with application of non-homogeneous magnetic field.
3. No effect was found in case of Zn, Cd, Hg, Sn and Pb.

# ***The Experiment:***

- **Setup:** In this experiment, silver atoms are passed through a non-uniform magnetic field.
- **Prediction by Classical Physics:** If angular momentum were continuous (as classical physics suggests), the atoms would spread out continuously on a screen after passing through the magnetic field.
- **Actual Observation:** Instead of spreading out, the atoms split into two distinct beams, hitting two specific spots on the screen.



## ***What This Means:***

- **Quantization:** This result showed that the angular momentum (spin) of particles like electrons is quantized; it can only take on certain discrete values.
- **Spin:** The experiment revealed that electrons (and other particles) have a property called spin, which can be thought of as a form of intrinsic angular momentum. For electrons, this spin can only be in one of two directions ("up" or "down").

1L

✓ 3.6: **Generalized uncertainty relation (Uncertainty principle for operators):**-Given two operators  $\hat{A}$  and  $\hat{B}$  such that  $[\hat{A}, \hat{B}] = i\hat{C}$ , the uncertainties  $\Delta\hat{A}$  and  $\Delta\hat{B}$  in an arbitrary state are related by,



$$\Delta\hat{A}\Delta\hat{B} \geq \frac{1}{2}|\langle\hat{C}\rangle|$$

Proof:-We know that,

$$(\Delta\hat{A})^2 = \langle\hat{A}^2\rangle - \langle\hat{A}\rangle^2$$

Let  $\hat{E} = \hat{A} - \langle\hat{A}\rangle$  and  $\hat{F} = \hat{B} - \langle\hat{B}\rangle$  and consider  $|\psi\rangle = (\hat{E} + i\lambda\hat{F})|\phi\rangle$  with  $\lambda$  real for simplicity. Then

$$\begin{aligned} \langle\psi, \psi\rangle &\geq 0 \Rightarrow \langle\phi|(\hat{E} - i\lambda\hat{F})(\hat{E} + i\lambda\hat{F})|\phi\rangle \geq 0 \\ \Rightarrow \langle\phi|(\hat{E}^2 + \lambda^2\hat{F}^2 + i\lambda(\hat{E}\hat{F} - \hat{F}\hat{E}))|\phi\rangle &\geq 0 \Rightarrow \langle\phi|(\hat{E}^2 + \lambda^2\hat{F}^2 + i\lambda[\hat{E}, \hat{F}])|\phi\rangle \geq 0 \\ \Rightarrow \langle\phi|((\hat{A} - \langle\hat{A}\rangle)^2 + \lambda^2(\hat{B} - \langle\hat{B}\rangle)^2 + i\lambda[\hat{A} - \langle\hat{A}\rangle, \hat{B} - \langle\hat{B}\rangle])|\phi\rangle &\geq 0 \\ \Rightarrow \langle\phi|((\Delta\hat{A})^2 + \lambda^2(\Delta\hat{B})^2 + i\lambda[\hat{A}, \hat{B}])|\phi\rangle &\geq 0 \rightarrow (3.2) \end{aligned}$$

The minimum will occur, when

$$\begin{aligned} \frac{d}{d\lambda} \{ \langle\phi|((\Delta\hat{A})^2 + \lambda^2(\Delta\hat{B})^2 + i\lambda[\hat{A}, \hat{B}])|\phi\rangle \} &= 0 \Rightarrow 2\lambda(\Delta\hat{B})^2 + i\langle\phi|[\hat{A}, \hat{B}]|\phi\rangle = 0 \\ \Rightarrow 2\lambda(\Delta\hat{B})^2 &= -i\langle\phi|[\hat{A}, \hat{B}]|\phi\rangle \Rightarrow \lambda = -\frac{i\langle\phi|[\hat{A}, \hat{B}]|\phi\rangle}{(\Delta\hat{B})^2} \end{aligned}$$

Putting value of  $\lambda$  in (3.2), we obtain

$$\begin{aligned} &(\Delta\hat{A})^2 - \frac{|\langle\phi|[\hat{A}, \hat{B}]|\phi\rangle|^2}{4(\Delta\hat{B})^4}(\Delta\hat{B})^2 + \frac{|\langle\phi|[\hat{A}, \hat{B}]|\phi\rangle|^2}{2(\Delta\hat{B})^2} \geq 0 \\ \Rightarrow (\Delta\hat{A})^2 - \frac{|\langle\phi|[\hat{A}, \hat{B}]|\phi\rangle|^2}{4(\Delta\hat{B})^2} + \frac{|\langle\phi|[\hat{A}, \hat{B}]|\phi\rangle|^2}{2(\Delta\hat{B})^2} &\geq 0 \Rightarrow (\Delta\hat{A})^2 + \frac{|\langle\phi|[\hat{A}, \hat{B}]|\phi\rangle|^2}{4(\Delta\hat{B})^2} \geq 0 \\ \Rightarrow (\Delta\hat{A})^2 &\geq -\frac{|\langle\phi|[\hat{A}, \hat{B}]|\phi\rangle|^2}{4(\Delta\hat{B})^2} \Rightarrow (\Delta\hat{A})^2(\Delta\hat{B})^2 \geq -\frac{|\langle\phi|[\hat{A}, \hat{B}]|\phi\rangle|^2}{4} \\ \Rightarrow (\Delta\hat{A})^2(\Delta\hat{B})^2 &\geq -\frac{|\langle\phi|i\hat{C}|\phi\rangle|^2}{4} \Rightarrow (\Delta\hat{A})^2(\Delta\hat{B})^2 \geq -\frac{i^2}{4}|\langle\phi|\hat{C}|\phi\rangle|^2 \\ \Rightarrow (\Delta\hat{A})^2(\Delta\hat{B})^2 &\geq \frac{1}{4}|\langle\hat{C}\rangle|^2 \Rightarrow \Delta\hat{A}\Delta\hat{B} \geq \frac{1}{2}|\langle\hat{C}\rangle|, \text{ which completes the proof.} \end{aligned}$$

The most general example is that of coordinate and canonical conjugate momentum, for which we have,

$$[\hat{x}, \hat{p}_x] = i\hbar$$

Then,

$$\Delta x \Delta p_x \geq \frac{\hbar}{2}, \quad \text{a familiar result.}$$

This shows that a particle with a well defined position cannot have well defined momentum.

2L

$$\Rightarrow H''(z) - 2zH'(z) + (\lambda - 1)H(z) = 0 \rightarrow (4.24)$$

This is Hermite differential equation. [See ILMI mathematical methods of physics ; engineering edition 2018 by same author page 329]

Solution of this equation will be of form,

$$H(z) = \sum_{n=0}^{\infty} a_n z^n, \quad H'(z) = \sum_{n=0}^{\infty} a_n n z^{n-1}, \quad H''(z) = \sum_{n=0}^{\infty} a_n n(n-1) z^{n-2}$$

Putting values in equation (4.24),

$$\sum_{n=0}^{\infty} a_n n(n-1) z^{n-2} - 2z \sum_{n=0}^{\infty} a_n n z^{n-1} + (\lambda - 1) \sum_{n=0}^{\infty} a_n z^n = 0$$

Put  $n \rightarrow n+2$

$$\Rightarrow \sum_{n=0}^{\infty} (n+1)(n+2) a_{n+2} z^n - \sum_{n=0}^{\infty} 2a_n n z^n + (\lambda - 1) \sum_{n=0}^{\infty} a_n z^n = 0$$

$$\Rightarrow \sum_{n=0}^{\infty} \{ (n+1)(n+2) a_{n+2} - 2a_n n + (\lambda - 1) a_n \} z^n = 0$$

$n+2 \rightarrow n$   
 $n+2-1 \rightarrow n-1$   
 $n+1$



$$\Rightarrow (n+1)(n+2)a_{n+2} - 2a_n n + (\lambda - 1)a_n = 0$$

$$\Rightarrow (n+1)(n+2)a_{n+2} - (2n - \lambda + 1)a_n = 0 \Rightarrow (n+1)(n+2)a_{n+2} = (2n - \lambda + 1)a_n$$

$$\Rightarrow a_{n+2} = \frac{(2n - \lambda + 1)}{(n+1)(n+2)} a_n ; n = 0, 1, 2, 3, \dots$$

This equation is called recursion formula.

For  $n = 0, 2, 4, \dots$

$$a_2 = \frac{(1 - \lambda)}{1 \cdot 2} a_0 = \frac{(1 - \lambda)}{2!} a_0$$

$$a_4 = \frac{(5 - \lambda)}{3 \cdot 4} a_2 = \frac{(5 - \lambda)(1 - \lambda)}{3 \cdot 4 \cdot 2!} a_0 = \frac{(5 - \lambda)(1 - \lambda)}{4!} a_0$$

$$a_6 = \frac{(9 - \lambda)}{5 \cdot 6} a_4 = \frac{(9 - \lambda)(5 - \lambda)(1 - \lambda)}{5 \cdot 6 \cdot 4!} a_0 = \frac{(9 - \lambda)(5 - \lambda)(1 - \lambda)}{6!} a_0 \text{ and so on.}$$

For  $n = 1, 3, 5, \dots$

$$a_3 = \frac{(3 - \lambda)}{2 \cdot 3} a_1 = \frac{(3 - \lambda)}{3!} a_1$$

$$a_5 = \frac{(7 - \lambda)}{4 \cdot 5} a_3 = \frac{(7 - \lambda)(3 - \lambda)}{4 \cdot 5 \cdot 3!} a_1 = \frac{(7 - \lambda)(3 - \lambda)}{5!} a_1$$

$$a_7 = \frac{(11 - \lambda)}{6 \cdot 7} a_5 = \frac{(11 - \lambda)(7 - \lambda)(3 - \lambda)}{6 \cdot 7 \cdot 5!} a_1 = \frac{(11 - \lambda)(7 - \lambda)(3 - \lambda)}{7!} a_1 \text{ and so on.}$$

Now,

$$H(z) = \sum_{n=0}^{\infty} a_n z^n = a_0 z^0 + a_1 z + a_2 z^2 + a_3 z^3 + a_4 z^4 + \dots$$

$$\Rightarrow H(z) = a_0 + a_1 z + \frac{(1 - \lambda)}{2!} a_0 z^2 + \frac{(3 - \lambda)}{3!} a_1 z^3 + \frac{(5 - \lambda)(1 - \lambda)}{4!} a_0 z^4 + \dots$$

$$\Rightarrow H(z) = a_0 \left\{ 1 + \frac{(1 - \lambda)}{2!} z^2 + \frac{(5 - \lambda)(1 - \lambda)}{4!} z^4 + \dots \right\} + a_1 \left\{ z + \frac{(3 - \lambda)}{3!} z^3 + \dots \right\}$$

This is solution of (4.24) which becomes infinite for infinite values of  $z$ . To obtain finite values of wave function and energy eigen values  $a_0$  must be zero for odd  $n$  and  $a_1$  must be zero for even  $n$ . Then for some value of  $n$ ,

$$a_{n+2} = 0 \Rightarrow \frac{(2n - \lambda + 1)}{(n+1)(n+2)} a_n = 0 \Rightarrow 2n - \lambda + 1 = 0 \Rightarrow \lambda = 2n + 1$$

Energy eigen values:- Putting this value of  $\lambda$  in  $2E/\hbar\omega = \lambda$ , we get

$$\frac{2E}{\hbar\omega} = 2n + 1 \Rightarrow E_n = \frac{1}{2} (2n + 1) \hbar\omega$$

$$\Rightarrow E_n = \left( n + \frac{1}{2} \right) \hbar\omega$$

For  $n = 0$ , energy is  $E_0 = \frac{\hbar\omega}{2}$ . This energy is called zero point energy, which is characteristic of quantum mechanics. Since energy of harmonic oscillator depends on  $n$  so all energy levels of oscillator are non-degenerate.

The successive energy levels are equally spaced and are shown in fig 4.19.

Wave functions:- The wave function of harmonic oscillator for state  $E_n$  is given by,

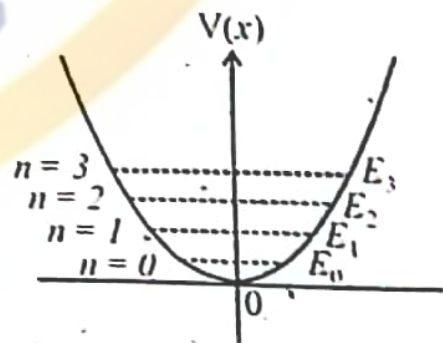


Fig 4.19: The energy levels for the lowest four energy states

$$\psi_n = a_n H_n(z) e^{-z^2/2}$$

Here  $a_n$  is a normalization constant which can be calculated by orthogonality condition of Hermite polynomials.

$$a_n = \left(\frac{\alpha}{\pi}\right)^{1/4} \frac{1}{\sqrt{2^n n!}} \quad \text{where } \alpha = \frac{m\omega}{\hbar}$$

So,

$$\psi_n = \left(\frac{\alpha}{\pi}\right)^{1/4} \frac{1}{\sqrt{2^n n!}} H_n(z) e^{-z^2/2} \quad \text{and } z = x\sqrt{\alpha} = x\sqrt{\frac{m\omega}{\hbar}}$$

Hence,

$$\psi_n(x) = \left(\frac{\alpha}{\pi}\right)^{1/4} \frac{1}{\sqrt{2^n n!}} H_n(x\sqrt{\alpha}) e^{-\alpha x^2/2}$$

It should be noted that  $H(z)$  are Hermite polynomials of degree  $n$ . Some Hermite polynomials are listed below:

$$H_0(z) = 1, H_1(z) = 2z, H_2(z) = 4z^2 - 1, H_3(z) = 8z^3 - 12z, H_4(z) = 16z^4 - 48z^2 + 12$$

For  $n = 0$ , wave function is given by,

$$\psi_0(x) = \left(\frac{\alpha}{\pi}\right)^{1/4} \frac{1}{\sqrt{2^0 0!}} H_0(x\sqrt{\alpha}) e^{-\alpha x^2/2} = \left(\frac{\alpha}{\pi}\right)^{1/4} e^{-\alpha x^2/2}$$

This wave function is called ground state wave function.

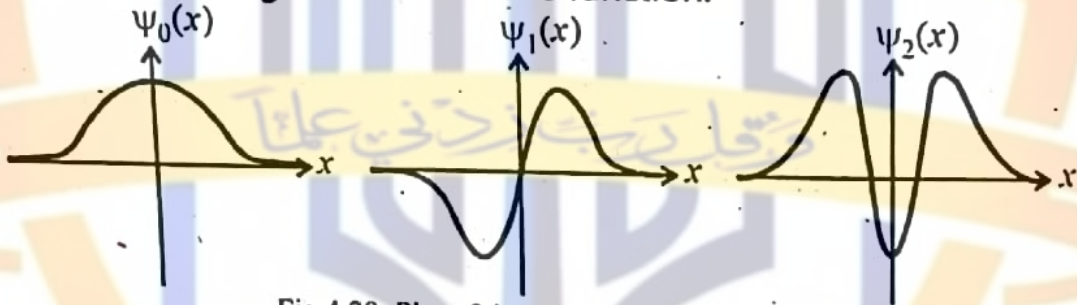


Fig 4.20: Plot of three lowest energy states

For  $n = 1$ , wave function is,

$$\begin{aligned} \psi_1(x) &= \left(\frac{\alpha}{\pi}\right)^{1/4} \frac{1}{\sqrt{2}} H_1(x\sqrt{\alpha}) e^{-\alpha x^2/2} \\ \Rightarrow \psi_1(x) &= \left(\frac{\alpha}{\pi}\right)^{1/4} \frac{1}{\sqrt{2}} \times 2x\sqrt{\alpha} e^{-\alpha x^2/2} \\ \Rightarrow \psi_1(x) &= \sqrt{2} \left(\frac{\alpha}{\pi}\right)^{1/4} x\sqrt{\alpha} e^{-\alpha x^2/2} \end{aligned}$$

The wave functions for the three lowest energy states are shown in fig 4.20. It should be noted that even quantum numbers have symmetric wave functions  $\psi_n(x)$  and the odd quantum numbers have anti-symmetric wave functions  $\psi_n(x)$ .

**Note:-**

The wave function is either even or odd depending on value of  $n$ .  
In fact  $\psi_{2n}(x)$  functions are even i.e.

$$\psi_{2n}(-x) = \psi_{2n}(x)$$

and  $\psi_{2n+1}(x)$  functions are odd i.e.

$$\psi_{2n+1}(-x) = -\psi_{2n+1}(x)$$

This is due to fact that Hermite polynomials  $H_{2n}(x)$  are even and  $H_{2n+1}(x)$  are odd.



3L

$$\hat{L}_+ \psi(l, m) = \sqrt{l(l+1) - m(m+1)} \hbar \psi(l, m+1)$$

$$\hat{L}_- \psi(l, m) = \sqrt{l(l+1) - m(m-1)} \hbar \psi(l, m-1)$$

**Proof:** Eigen value equations for  $\hat{L}_z$  and  $\hat{L}^2$  are,

$$\hat{L}_z \psi(l, m) = m \hbar \psi(l, m) \quad \text{and} \quad \hat{L}^2 \psi(l, m) = l(l+1) \hbar^2 \psi(l, m)$$

Operating first equation by  $\hat{L}_+$ ,

$$\begin{aligned} \hat{L}_+ \hat{L}_z \psi(l, m) &= m \hbar \hat{L}_+ \psi(l, m) \Rightarrow (\hat{L}_z \hat{L}_+ - \hbar \hat{L}_+) \psi(l, m) = m \hbar \hat{L}_+ \psi(l, m) \\ \Rightarrow \hat{L}_z \hat{L}_+ \psi(l, m) - \hbar \hat{L}_+ \psi(l, m) &= m \hbar \hat{L}_+ \psi(l, m) \Rightarrow \hat{L}_z \hat{L}_+ \psi(l, m) = \hbar \hat{L}_+ \psi(l, m) + m \hbar \hat{L}_+ \psi(l, m) \\ \Rightarrow \hat{L}_z \hat{L}_+ \psi(l, m) &= (m+1) \hbar \hat{L}_+ \psi(l, m) \end{aligned}$$

Multiplying both sides with C,

$$\hat{L}_z \{C \hat{L}_+ \psi(l, m)\} = (m+1) \hbar \{C \hat{L}_+ \psi(l, m)\} \rightarrow (a)$$

Thus  $\{C \hat{L}_+ \psi(l, m)\}$  is an eigen function of  $\hat{L}_z$  corresponding to eigen value  $(m+1)\hbar$ .

Now,

$$\begin{aligned} \hat{L}_+ \hat{L}^2 \psi(l, m) &= l(l+1) \hbar^2 \hat{L}_+ \psi(l, m) \\ \Rightarrow \hat{L}^2 \hat{L}_+ \psi(l, m) &= l(l+1) \hbar^2 \hat{L}_+ \psi(l, m) \quad \because [\hat{L}_+, \hat{L}^2] = 0 \end{aligned}$$

Multiplying both sides with C,

$$\hat{L}^2 \{C \hat{L}_+ \psi(l, m)\} = l(l+1) \hbar^2 \{C \hat{L}_+ \psi(l, m)\} \rightarrow (b)$$

From equations (a) and (b), one can observe that  $l$  remains same but  $m$  increases by 1.

Hence we can put,

$$C \hat{L}_+ \psi(l, m) = \psi(l, m+1) \Rightarrow \hat{L}_+ \psi(l, m) = \frac{1}{C} \psi(l, m+1) = A \psi(l, m+1) \quad \text{where } \frac{1}{C} = A$$

$$\hat{L}_+ \psi(l, m) = A \psi(l, m+1) \rightarrow (c)$$

Taking complex conjugate,

$$\{\hat{L}_+ \psi(l, m)\}^* = A^* \psi^*(l, m+1) \rightarrow (d)$$

Multiplying equations (c) and (d) and integrating,

$$\begin{aligned} \int_{-\infty}^{\infty} \{\hat{L}_+ \psi(l, m)\}^* \hat{L}_+ \psi(l, m) d\tau &= \int_{-\infty}^{\infty} A^* \psi^*(l, m+1) A \psi(l, m+1) d\tau \\ \Rightarrow \int_{-\infty}^{\infty} \psi^*(l, m) \hat{L}_- \hat{L}_+ \psi(l, m) d\tau &= |A|^2 \int_{-\infty}^{\infty} \psi^*(l, m+1) \psi(l, m+1) d\tau \end{aligned}$$

$$\Rightarrow \int_{-\infty}^{\infty} \psi^*(l, m) (\hat{L}^2 - \hat{L}_z^2 - \hbar \hat{L}_z) \psi(l, m) d\tau = |A|^2$$

$$\Rightarrow |A|^2 = \int_{-\infty}^{\infty} \psi^*(l, m) \{l(l+1) \hbar^2 - m^2 \hbar^2 - m \hbar^2\} \psi(l, m) d\tau$$

$$\Rightarrow |A|^2 = \{l(l+1) \hbar^2 - m^2 \hbar^2 - m \hbar^2\} \int_{-\infty}^{\infty} \psi^*(l, m) \psi(l, m) d\tau$$

$$\Rightarrow |A|^2 = l(l+1) \hbar^2 - m(m+1) \hbar^2 \Rightarrow A = \sqrt{l(l+1) - m(m+1)} \hbar$$

Putting value of A in (c), we get

$$\hat{L}_+ \psi(l, m) = \sqrt{l(l+1) - m(m+1)} \hbar \psi(l, m+1)$$

Similarly it can be proved that,

$$\hat{L}_- \psi(l, m) = \sqrt{l(l+1) - m(m-1)} \hbar \psi(l, m-1)$$

✓ proved